

Vinay Jampana

Senior Software Engineer – AI & Full-Stack

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Summary

AI full-stack engineer with 5 years across React/TypeScript, Node/NestJS and AWS. I design and ship production AI agents end to end, with the evaluation and tracing that keep them reliable. AI engineering since 2024.

Skills

AI: LLM agents, tool calling, multi-agent, LangGraph, RAG and vector search (OpenSearch), evaluation, tracing (OpenTelemetry, Langfuse)

Backend: Python, FastAPI, Node.js, NestJS, PostgreSQL

Frontend: React, Next.js, TypeScript

Cloud: AWS, Docker, GitHub Actions

Experience

Zotok.ai – Senior Software Engineer (AI & Full-Stack)

May 2023 – Present

Stack: LangGraph, Python, TypeScript, NestJS, PostgreSQL, OpenSearch, React, Next.js, AWS

- Designed a configurable WhatsApp order agent (**LangGraph**): one black box became **seven testable steps** on a visual builder, so per-customer changes ship without code or a redeploy; running for multiple B2B customers
- Designed the **multi-agent** layer of the WhatsApp sales agent (**LangGraph**): messages route between service and payment agents, and a **guardrail classifier** filters spam, auto-replies and off-topic chat before the model answers
- Built a **product-matching engine** for shorthand order text over **15,242 production aliases**: the correct first match rose from **69% to 78%** in testing and from **81% to 95%** on real rep messages
- Built the **evaluation system** for the order agent: **64 real scenarios** graded line by line against the correct products, run **twice daily** with Teams alerts; it repeatedly caught breakages from other teams' backend changes before production
- Built end-to-end **tracing (OpenTelemetry, Langfuse)** across every agent step, so a failed AI order is debugged on one screen instead of in server logs
- Built the agent's **knowledge search (OpenSearch vector search, LiteLLM embeddings)** with per-customer isolation, so every customer's answers come only from their own documents
- Built **image-to-order**: photos of handwritten order sheets become carts through **OCR** and product matching, at **88% line accuracy** on a 378-line test set
- Built **LLM cost tracking** on a **LiteLLM** gateway: per-workspace keys and spend tags per feature, so model cost is attributable by customer and feature
- Built end to end in **2.5 weeks**: a **WhatsApp** workflow (agent, **NestJS** backend, **React** web app) that replaced a plant's offline pour and dispatch forms; over **3,800 pour cards** and **1,900 dispatches** captured across 200 projects since launch
- Built the order service behind the agent (**NestJS, PostgreSQL**): over **50 typed functions** on one shared contract, with product search **4x faster** (875 to 228 ms per line)
- Built **multi-address ordering**: one buyer message becomes one order per delivery location, with any partial failure reported back to the buyer
- Cut first page load from **15–30 seconds to about 1.5 seconds** by removing unused mobile code, rebuilding the **Webpack** config and lazy loading; builds got **30–40% faster**

Sumeru Software – Senior Developer

Aug 2022 – Jan 2023

- Built a **role-based CRM** from scratch (**React**, Redux Toolkit, **Node.js**) with permission-based screens and modular dashboards; owned it from kickoff to production and shipped the **MVP in 4 months** with a team of 4

Aerchain – Software Development Engineer

Intern Jul 2019 – Jul 2020; SDE 1 Apr 2021 – Jan 2022

- Built the **real-time auction** module for a B2B procurement platform, with an **optimistic UI** that reconciles with the server so bids feel instant and stay correct
- Built barcode-scan inventory management as an **offline-first PWA** (service workers) that keeps working without a connection

Education

B.E. (Hons) Mechanical Engineering, BITS Pilani

2016 – 2020